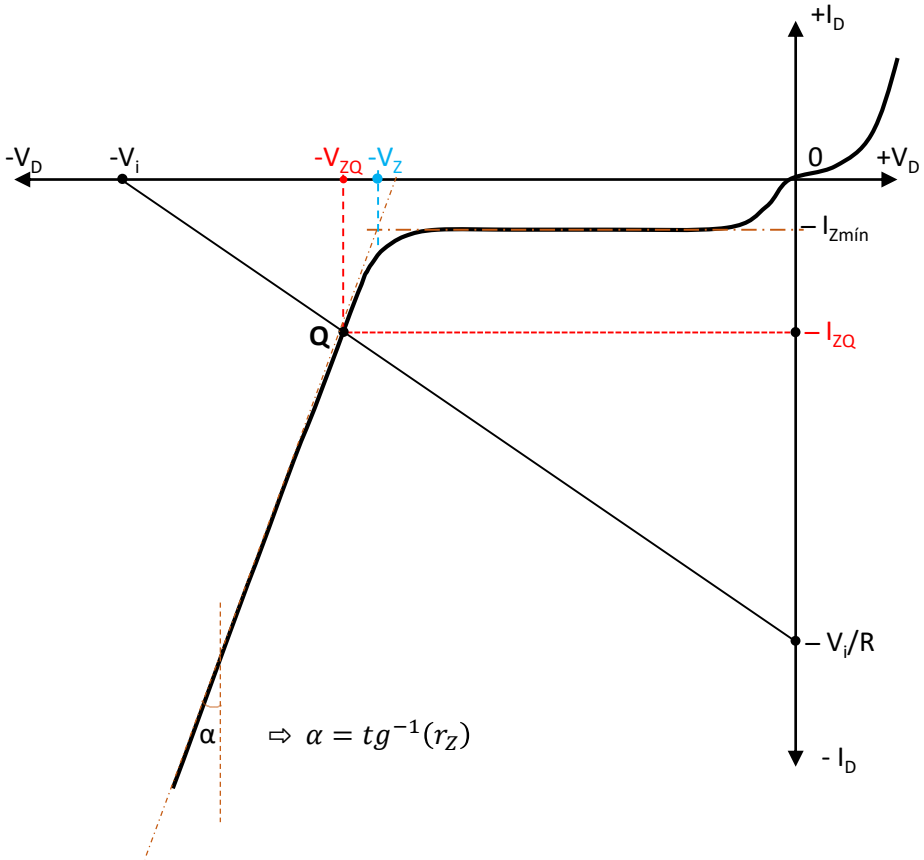
Introducción a la Electrónica

Unidades 1 y 3

Diodo Zener + TBJ

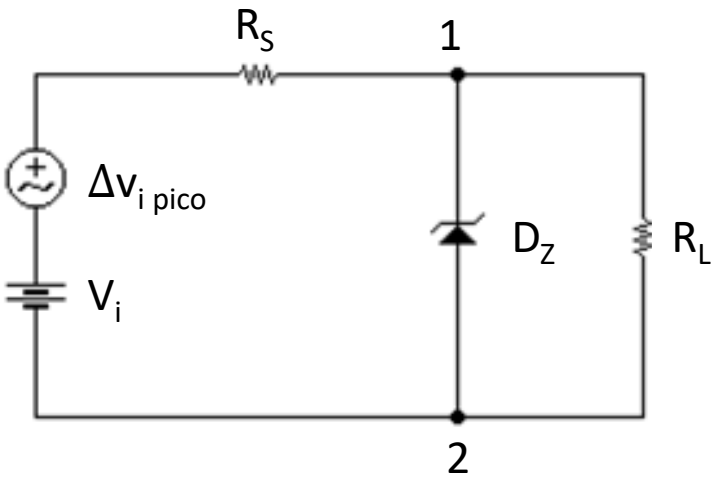
Dr. Ing. David M. Petruzzi



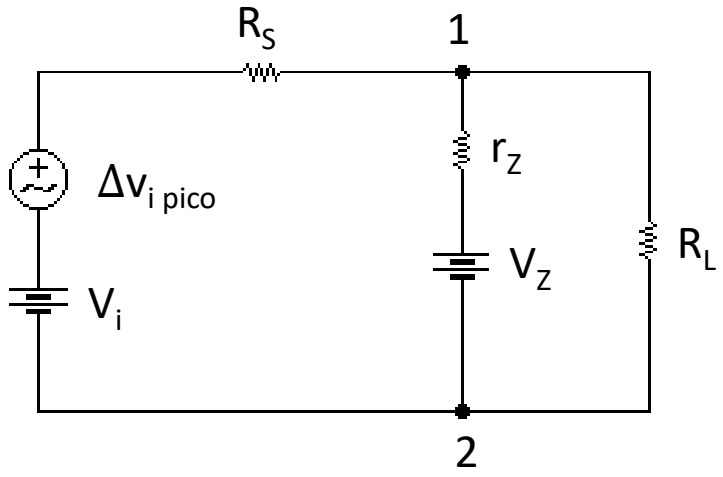
$$V_{ZQ} = V_Z + I_{ZQ} . r_Z$$

$$I_{ZQ} > I_{Zmín}$$

Supongamos tener una fuente de alimentación, a partir de un circuito rectificador con filtrado capacitivo. El mismo entrega V_i Volts de CC y produce una tensión de zumbido de ΔV_i Volts pico:



circuito equivalente
(aproximación por tramos)

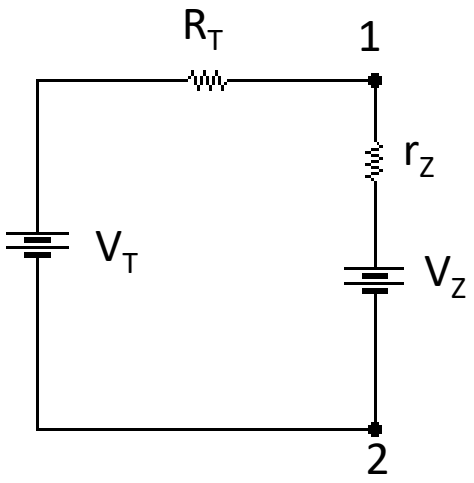


Realizamos ahora un análisis, separando la CC de la CA

Thévenin correspondiente a CC:

$$R_T = R_S \| R_L$$

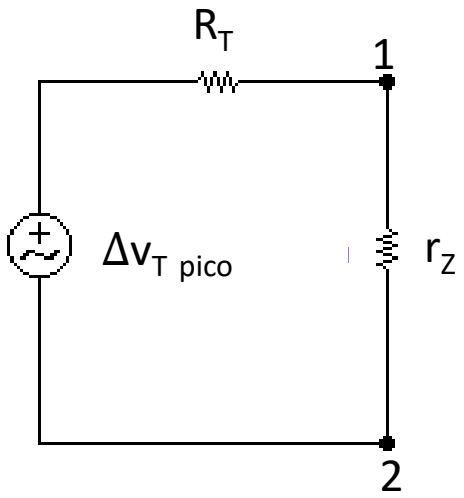
$$V_T = V_i \cdot \left(\frac{R_L}{R_S+R_L}\right)$$



Thévenin correspondiente a CA:

$$R_T = R_S \| R_L$$

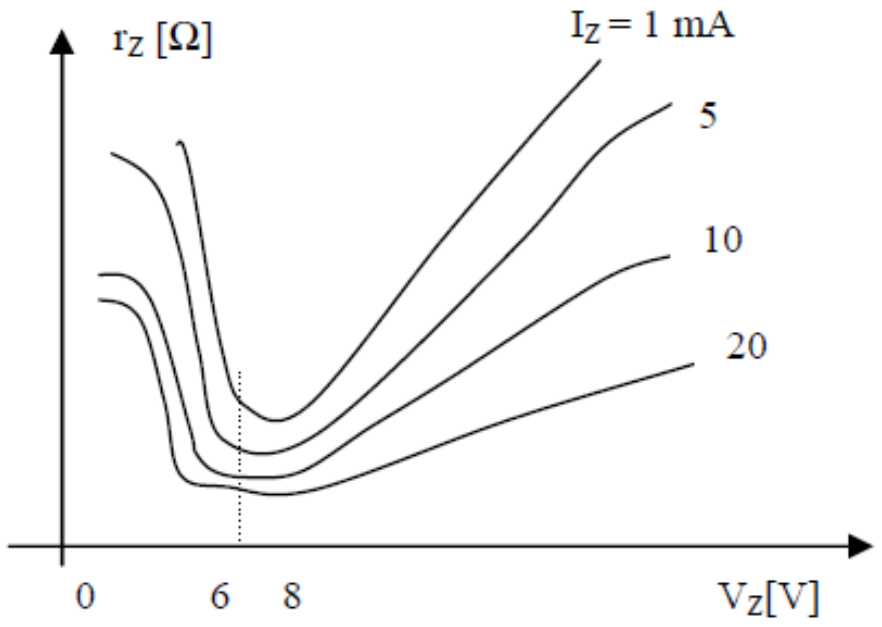
$$\Delta V_T = \Delta V_i \cdot \left(\frac{R_L}{R_S+R_L}\right)$$

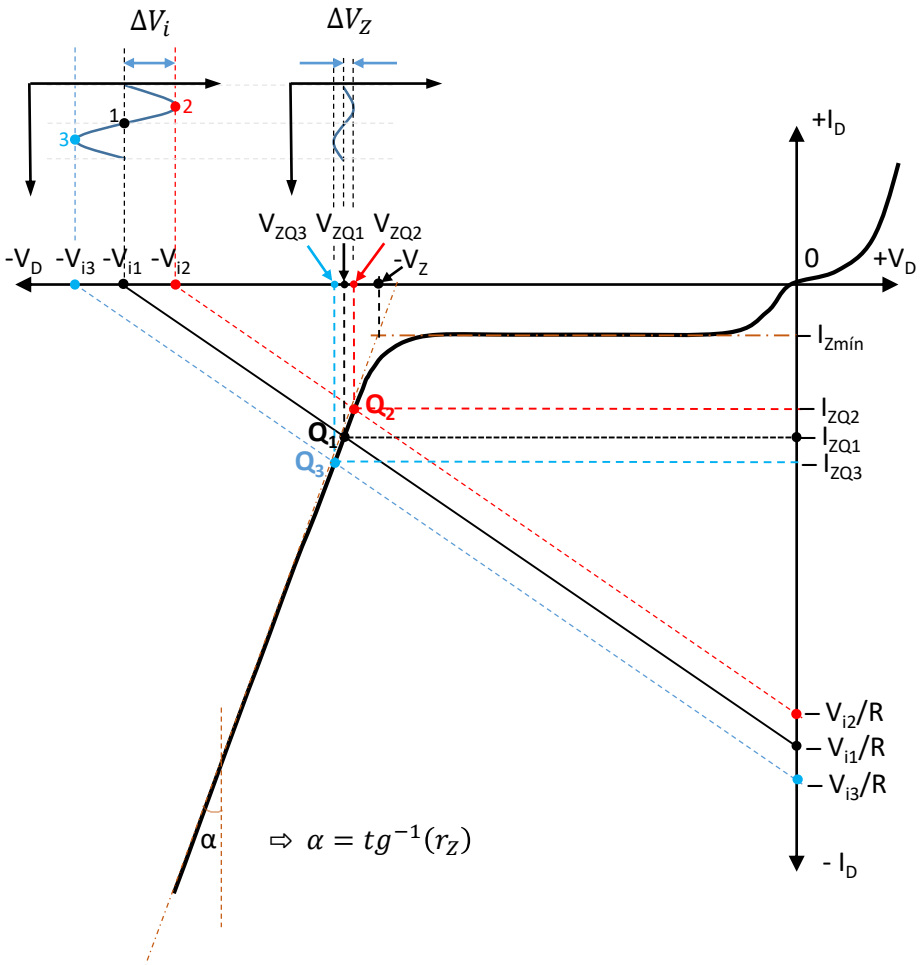


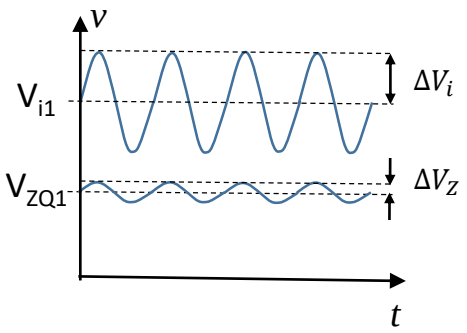
donde : $r_z = dV / dI$ y es una resistencia *dinámica*

y $V_Z \equiv$ Tensión de ruptura en inversa

Nota: El valor de la resistencia dinámica r_Z depende del parde valores: V_Z e I_Z (punto sobre la curva).







$$V_{ZQ} = V_Z + I_{ZQ} . r_Z$$

$$\Delta V_Z = \Delta V_i . \left[\frac{r_Z}{r_Z + R}\right]$$

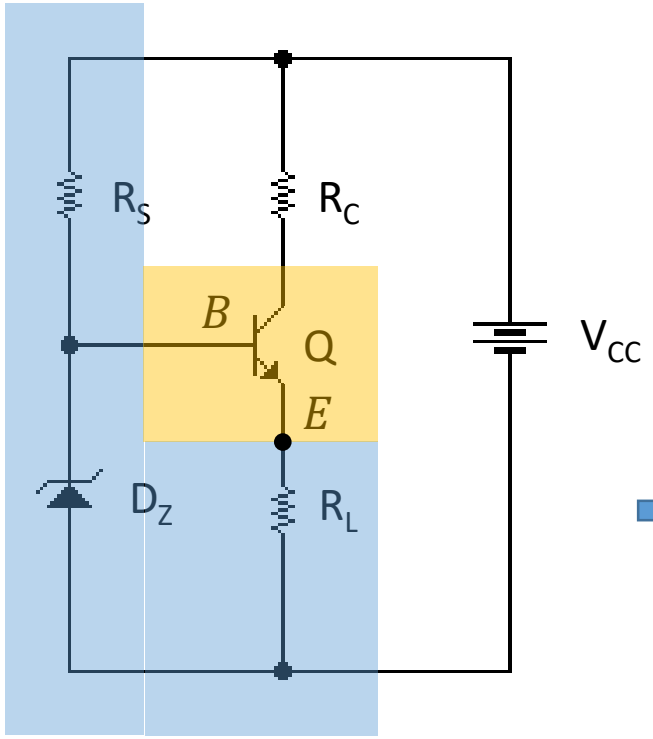
si: $r_Z \ll R \Rightarrow \Delta V_Z \cong \Delta V_i . \frac{r_Z}{R}$

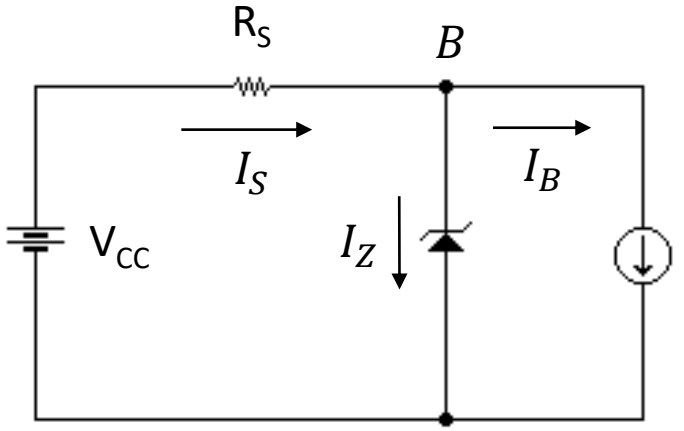
$$P_d = P_{CC} + P_{CA} = V_{ZQ} . I_{ZQ} + \frac{(\Delta V_Z)^2}{k . r_Z} =$$

$$= V_Z . I_{ZQ} + I_{ZQ}{}^2 . r_Z + \frac{(\Delta V_Z)^2}{k . r_Z}$$

$$k = \begin{cases} 2 \Rightarrow \textit{onda senoidal} \\ 3 \Rightarrow \textit{diente de sierra} \end{cases}$$

Intercalando un TBJ entre D_Z y la carga, como muestra el sgte. circuito, mejoramos la Regulación de la Tensión V_Z para grandes cambios de R_L (el punto Q se desplaza poco)





$$I_B = \frac{V_{ZQ} - V_\gamma}{(\beta + 1)R_L} = \frac{V_{R_L}}{(\beta + 1)R_L}$$

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$$I_B = \frac{I_E}{(\beta+1)} = \frac{V_{R_L}}{(\beta+1)R_L} = \frac{V_{ZQ} - V_\gamma}{(\beta+1)R_L}$$

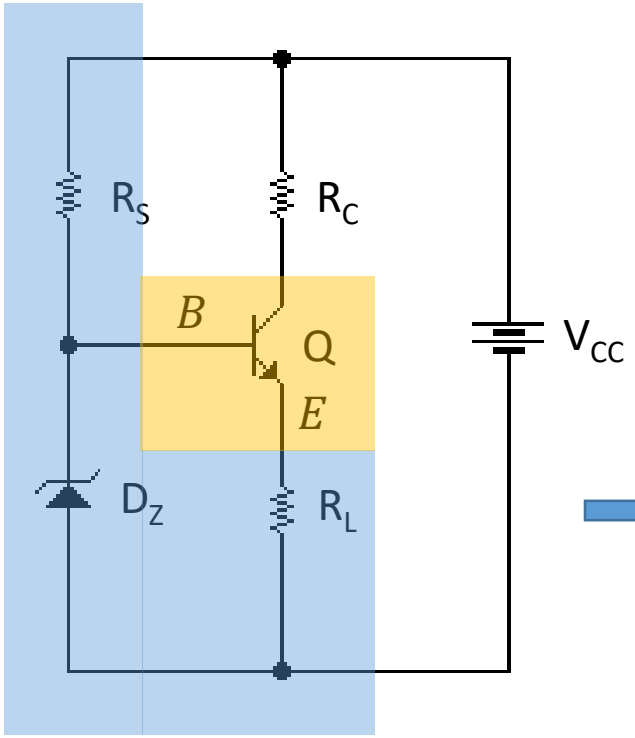
$[(\beta+1).R_L]$ debe ser lo suficientemente grande, como para garantizar que $I_B \ll I_S$

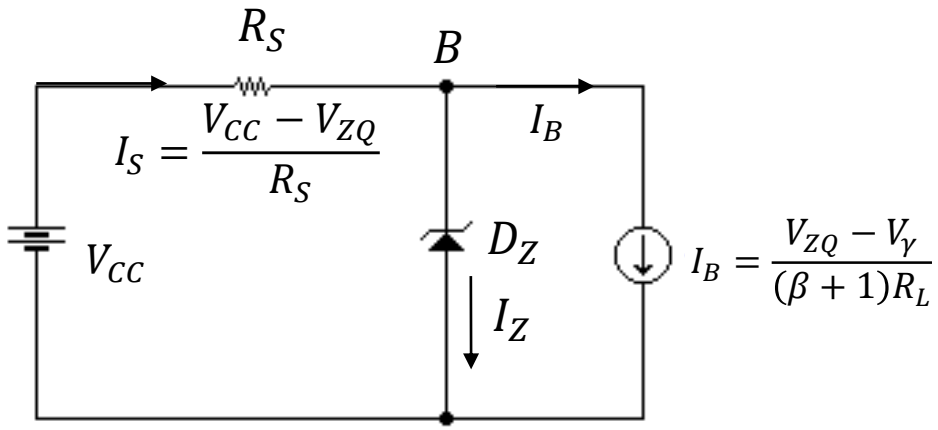
$$I_B \ll I_S \Longrightarrow I_S \cong I_Z$$

$$\frac{V_{ZQ} - V_\gamma}{(\beta+1)R_L} \ll \frac{V_{CC} - V_{ZQ}}{R_S}$$

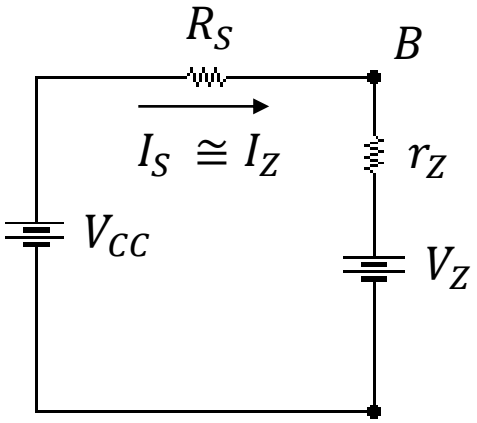
$$R_T \cong R_S$$

$$V_T \cong V_{CC}$$

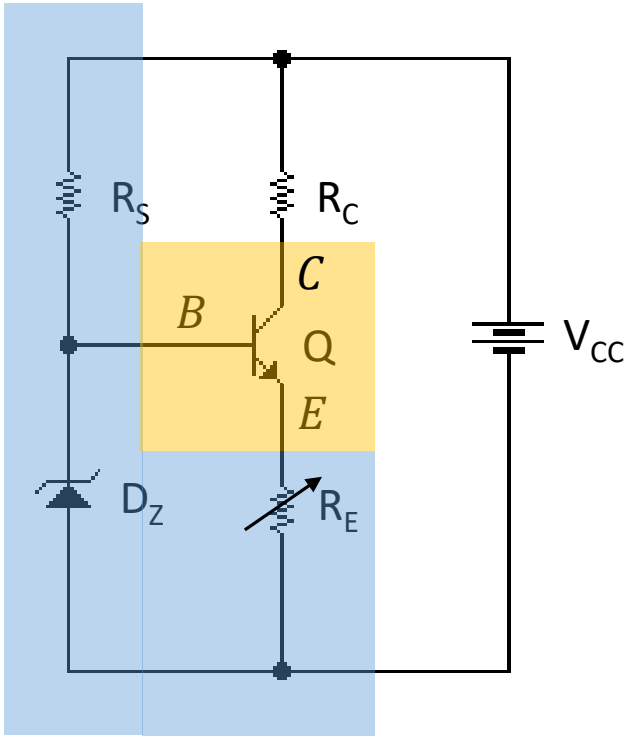




$\cong$



Fuente de corriente ajustable utilizando:
Diodo Zener + TBJ + Potenciómetro Lineal



$$I_C \cong \frac{V_{ZQ} - V_\gamma}{R_E}$$

$$I_{C_{máx}}.\left(R_C + R_{E_{mín}}\right) \leq V_{CC} - V_{CE_{sat}}$$